

PISA GALILEO GALILEI, Italy

WMO: 161580

Lat: 43.6823N

Lon: 10.3955E

Elev: 6

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-0.9	-10.7	1.5	4.4	-8.1	1.9	4.7	11.2	11.4	10.0	10.5	2.4	110	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	32.2	21.8	31.1	21.8	30.0	21.5	24.8	28.6	24.0	27.8	23.3	27.2	4.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	18.7	26.8	22.9	17.7	26.2	22.0	16.6	25.5	75.4	28.8	72.0	27.9	69.0	27.2	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.4	7.2	DB	-4.6	35.0	1.3	1.5	-5.5	36.0	-6.3	36.9	-7.1	37.7	-8.0	38.8	(4)
			WB	-5.2	26.4	1.4	1.6	-6.2	27.6	-6.9	28.5	-7.7	29.4	-8.7	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	7.1	7.7	10.3	13.1	16.9	21.1	23.9	24.2	20.3	16.5	12.0	8.0	(6)
		DBStd	6.59	2.96	2.86	2.46	2.38	2.25	2.64	2.01	2.06	2.46	2.76	3.44	3.39	(7)
		HDD10.0	299	98	72	26	3	0	0	0	0	1	21	80		(8)
		HDD18.3	1699	348	297	249	157	55	5	0	9	68	192	320		(9)
		CDD10.0	2178	8	8	36	97	216	334	430	439	311	203	79	18	(10)
		CDD18.3	537	0	0	0	0	12	89	172	182	69	13	1	0	(11)
		CDH23.3	4674	0	0	0	3	110	789	1655	1649	434	34	0	0	(12)
		CDH26.7	1501	0	0	0	0	13	229	585	593	80	1	0	0	(13)
(14)	Wind	WSAvg	2.9	2.9	3.0	3.1	2.9	2.7	2.7	2.8	2.7	2.8	2.8	3.1	3.1	(14)
(15)	Precipitation	PrecAvg	885	74	68	72	73	62	44	23	51	91	128	128	86	(15)
(16)		PrecMax	1470	239	208	241	251	180	116	84	246	319	383	374	199	(16)
(17)		PrecMin	510	0	3	0	2	2	0	0	0	0	6	0	0	(17)
(18)		PrecStd	195	52	44	50	52	49	28	22	49	64	95	77	45	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	15.2	16.1	20.0	23.8	28.2	32.9	33.9	34.6	30.8	25.9	20.8	16.9		(19)
(20)		0.4% MCWB	11.8	10.7	12.6	15.9	18.7	22.8	21.4	21.5	20.1	18.1	17.2	13.9		(20)
(21)		2% DB	14.1	14.9	17.8	21.8	26.1	31.0	32.1	32.3	28.9	24.2	19.2	15.2		(21)
(22)		2% MCWB	11.5	10.5	11.7	14.6	17.7	21.8	21.3	22.0	20.0	18.5	15.9	12.8		(22)
(23)		5% DB	13.1	13.9	16.6	20.0	24.2	29.2	31.0	31.1	27.2	22.9	18.1	14.2		(23)
(24)		5% MCWB	10.5	10.3	11.6	14.1	17.2	20.8	21.7	22.0	19.8	18.3	15.0	11.9		(24)
(25)		10% DB	12.1	12.9	15.2	18.3	22.8	27.8	29.9	29.9	25.9	21.6	17.0	13.2		(25)
(26)		10% MCWB	9.7	9.8	11.0	13.5	16.7	20.5	21.5	21.9	19.4	17.4	14.4	11.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	13.2	13.0	14.6	17.3	20.5	25.5	25.6	26.0	23.7	21.3	18.3	14.7		(27)
(28)		0.4% MCDB	14.6	14.3	17.1	21.0	25.1	29.6	29.1	29.0	26.9	23.3	19.8	16.0		(28)
(29)		2% WB	12.1	11.9	13.4	16.0	19.3	23.7	24.6	25.1	22.6	20.3	16.6	13.6		(29)
(30)		2% MCDB	13.8	13.7	15.9	19.5	23.8	28.2	28.4	28.6	25.7	22.4	18.5	15.0		(30)
(31)		5% WB	11.2	11.1	12.7	15.1	18.5	22.5	23.8	24.2	21.6	19.3	15.7	12.6		(31)
(32)		5% MCDB	12.9	13.0	15.1	18.5	22.7	27.0	27.8	27.8	25.1	21.5	17.5	14.0		(32)
(33)		10% WB	10.3	10.4	12.0	14.2	17.6	21.4	23.1	23.4	20.7	18.3	14.8	11.5		(33)
(34)		10% MCDB	11.7	12.1	14.4	17.4	21.4	25.8	27.2	27.2	24.2	20.6	16.6	12.9		(34)
(35)	Mean Daily Temperature Range	MDBR	7.9	8.7	9.3	9.7	10.2	10.8	11.2	10.9	10.3	9.0	7.5	7.5		(35)
(36)		5% DB MCDBR	8.2	10.1	11.5	12.2	13.0	13.0	12.9	12.3	11.9	10.1	7.8	7.5		(36)
(37)		5% WB MCWBR	5.9	6.7	6.9	6.7	6.5	5.9	5.6	5.5	5.6	5.6	5.0	5.3		(37)
(38)		5% WB MCDBR	6.9	7.9	9.3	10.5	11.2	11.6	10.9	10.7	9.8	8.0	6.6	5.8		(38)
(39)		5% WB MCWBR	5.6	6.1	6.3	6.5	6.3	6.1	5.6	5.7	5.6	5.3	4.9	4.8		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.356	0.391	0.423	0.430	0.435	0.428	0.422	0.408	0.393	0.364	0.334		(40)
(41)		taud	2.468	2.397	2.306	2.232	2.260	2.292	2.311	2.345	2.364	2.390	2.412	2.484		(41)
(42)		Ebn at Noon	789	831	843	842	846	840	841	834	817	777	740	754		(42)
(43)		Edh at Noon	73	92	114	133	133	129	126	118	107	92	77	67		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.53	2.37	3.63	4.92	6.11	6.97	7.15	6.17	4.54	2.83	1.67	1.32		(44)
(45)		RadStd	0.17	0.29	0.49	0.41	0.51	0.37	0.36	0.34	0.22	0.31	0.18	0.15		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+64	N/A

Nomenclature: See separate page